

Spider-Eyes GD-FMM: A Deterministic Multi-View Ensemble Specification for RNA 3D Folding (Five-Structure Output)

1 Purpose

This document specifies a mathematically deterministic “Spider-Eyes” architecture that produces an ensemble of E candidate RNA 3D structures from a single RNA sequence. For Kaggle-style evaluation requiring multiple candidate structures, we fix $E = 5$.

The central idea is:

- Each *eye* computes a distinct internal representation (a *mosaic*) of the same RNA input.
- Each eye outputs a discrete set of geometric constraints via categorical “dice” distributions.
- Each eye deterministically reconstructs one 3D coordinate set from those constraints.
- A Mosaic-of-Mosaics (MoM) computes cross-eye coherence and novelty diagnostics, and may deterministically reweight/spawn/merge eyes (optional), while still outputting five structures.

2 Sets, Indices, and Notation

Let the RNA sequence length be $L \in \mathbb{N}$ with residues indexed by $r \in \{1, \dots, L\}$. The nucleotide alphabet is $\Sigma = \{A, C, G, U\}$.

We output E structures, indexed by $e \in \{1, \dots, E\}$ with $E = 5$ for submission.

We define a set of residue-pair constraints

$$P = \{(a_k, b_k)\}_{k=1}^N, \quad 1 \leq a_k < b_k \leq L,$$

where each pair k corresponds to one “die.”

Each die has F discrete faces (distance bins), indexed by $f \in \{1, \dots, F\}$.

The shared context dimension is D , and each eye produces a mosaic of dimension H .

Vectors are column vectors by default; $\odot$ denotes elementwise multiplication; $\|\cdot\|_2$ is Euclidean norm.

3 Spider-Eyes as Multi-Projection Encoding

Define the set of projection operators (eyes):

$$\mathcal{E} = \{\Pi_1, \Pi_2, \dots, \Pi_E\}.$$

Each eye Π_e maps the same input context to a distinct mosaic and then to dice distributions.

3.1 Required properties

1. **Distinctness across eyes.** For typical inputs x , mosaics should differ:

$$m^{(e)}(x) \neq m^{(e')}(x) \quad \forall e \neq e'.$$

Distinctness is enforced by eye-specific parameters and an explicit novelty mechanism in MoM (Section 7).

2. **Class-preserving.** Residue order and identity are preserved by construction because dice indices k are tied to explicit residue pairs (a_k, b_k) and the encoder retains positional structure.
3. **Similarity-distorting.** Eyes differ in gating angles and temperature, inducing different internal geometries that promote exploration of alternative conformational modes.

4 Deterministic RNA Input Encoding

4.1 Per-residue features

For residue r , define a deterministic feature vector $u_r \in \mathbb{R}^{d_u}$:

$$u_r = [\text{onehot}(s_r); \text{pos}(r); (\text{optional fixed extras})],$$

where $\text{onehot}(s_r) \in \{0, 1\}^4$ is the nucleotide one-hot encoding and $\text{pos}(r)$ is a deterministic positional encoding.

4.2 Global context

Define a global context vector $x \in \mathbb{R}^D$:

$$x = \phi \left(\frac{1}{L} \sum_{r=1}^L W_u u_r + b_u \right),$$

where $W_u \in \mathbb{R}^{D \times d_u}$, $b_u \in \mathbb{R}^D$, and ϕ is a fixed nonlinearity (e.g. $\tanh$).

5 Dice Definition for RNA Geometry

We define dice over discretized residue-pair distances.

5.1 Pair set P

Choose P deterministically. One valid parameterization uses window W and stride S :

- Backbone neighbors: $(r, r + 1)$ for $r = 1, \dots, L - 1$.
- Local window: $(r, r + \Delta)$ for $\Delta = 2, \dots, W$ where defined.
- Long-range sparse pairs: $(r, r + tS)$ for integer t with $r + tS \leq L$, optionally for r in a fixed subset (e.g. $r \equiv 1 \pmod{S}$).

5.2 Distance bins

Let bin edges satisfy

$$0 = b_0 < b_1 < \dots < b_F = d_{\max},$$

and define representative bin distances

$$\bar{d}_f = \frac{b_{f-1} + b_f}{2}.$$

Die face f represents distance in $[b_{f-1}, b_f)$.

6 Eye Architecture (Gate $\rightarrow$ Cluster $\rightarrow$ Mosaic $\rightarrow$ Dice)

Each eye e has parameters:

- Gate angles $A^{(e)} \in \mathbb{R}^{G \times D}$ (with G gates),
- Cluster weights and biases $\{W_{c,j}^{(e)}, b_{c,j}^{(e)}\}_{j=1}^C$,
- Dice head weights and biases $\{w_{k,f}^{(e)}, b_{k,f}^{(e)}\}$,
- Temperature $\tau_e > 0$.

6.1 Gate layer

For gate $g \in \{1, \dots, G\}$:

$$z_g^{(e)}(x) = \tanh(A_g^{(e)} \odot x) \in \mathbb{R}^D,$$

and the aggregated gate output is

$$g^{(e)}(x) = \frac{1}{G} \sum_{g=1}^G z_g^{(e)}(x) \in \mathbb{R}^D.$$

6.2 Cluster layer

Let there be C clusters, each outputting a vector in $\mathbb{R}^h$:

$$c_j^{(e)}(x) = \tanh(W_{c,j}^{(e)} g^{(e)}(x) + b_{c,j}^{(e)}) \in \mathbb{R}^h,$$

where $W_{c,j}^{(e)} \in \mathbb{R}^{h \times D}$ and $b_{c,j}^{(e)} \in \mathbb{R}^h$.

6.3 Mosaic

Concatenate cluster outputs into a mosaic:

$$m^{(e)}(x) = \text{concat}(c_1^{(e)}(x), \dots, c_C^{(e)}(x)) \in \mathbb{R}^H, \quad H = C \cdot h.$$

6.4 Dice logits and categorical distributions

For each die $k \in \{1, \dots, N\}$ and face $f \in \{1, \dots, F\}$:

$$\ell_{k,f}^{(e)}(x) = \langle w_{k,f}^{(e)}, m^{(e)}(x) \rangle + b_{k,f}^{(e)}.$$

Define the per-die distribution via temperature-scaled softmax:

$$p_{k,f}^{(e)}(x) = \frac{\exp(\ell_{k,f}^{(e)}(x)/\tau_e)}{\sum_{f'=1}^F \exp(\ell_{k,f'}^{(e)}(x)/\tau_e)}.$$

Thus each eye yields $P^{(e)}(x) = \{p_k^{(e)}(x)\}_{k=1}^N$.

7 MoM: Mosaic-of-Mosaics (Coherence and Novelty)

MoM computes cross-eye disagreement (coherence) and diversity (novelty). These can be used as diagnostics, for deterministic weighting, or for optional spawn/merge rules.

7.1 Jensen–Shannon divergence

For categorical distributions $p, q \in \Delta^{F-1}$:

$$\text{JS}(p, q) = \frac{1}{2} \text{KL}(p \| m) + \frac{1}{2} \text{KL}(q \| m), \quad m = \frac{p + q}{2},$$

where

$$\text{KL}(p \| m) = \sum_{f=1}^F p_f \log \frac{p_f}{m_f}.$$

A fixed $\epsilon > 0$ may be added for numerical stability but must be constant.

7.2 Disagreement

Define mean disagreement across dice and eye pairs:

$$\text{Disagree}(x) = \frac{1}{N} \sum_{k=1}^N \frac{2}{E(E-1)} \sum_{1 \leq e < e' \leq E} \text{JS}(p_k^{(e)}(x), p_k^{(e')}(x)).$$

7.3 Behavior vectors and novelty

Define a deterministic behavior vector by compressing the mosaic into K chunk means and L2 normalizing:

$$b^{(e)}(x) = \text{Normalize}\left(\text{ChunkMean}_K(m^{(e)}(x))\right) \in \mathbb{R}^K.$$

MoM maintains an archive $A \subset \mathbb{R}^K$ of past behavior vectors (deterministic update; bounded capacity).

Novelty is defined as mean distance to the archive:

$$\text{Novel}(x) = \begin{cases} \frac{1}{E|A|} \sum_{e=1}^E \sum_{a \in A} \|b^{(e)}(x) - a\|_2, & |A| > 0 \\ \frac{1}{E} \sum_{e=1}^E \|b^{(e)}(x)\|_2, & |A| = 0. \end{cases}$$

7.4 Optional task proxy

During inference, the true Kaggle score is unavailable, so a deterministic *proxy* may be used:

$$\text{Task}(x) = \frac{1}{E} \sum_{e=1}^E \text{Task}_e(X^{(e)}),$$

where $X^{(e)}$ are reconstructed coordinates (Section 8). Examples include penalties for chain discontinuity, excessive curvature, or coarse clash counts.

7.5 MoM energy

Define the MoM energy:

$$\mathcal{E}(x) = \alpha \cdot \text{Disagree}(x) - \beta \cdot \text{Novel}(x) + \gamma \cdot \text{Task}(x),$$

with fixed $\alpha, \beta, \gamma \geq 0$.

7.6 Deterministic archive update (capacity M)

Let per-eye novelty score be $n_e = \min_{a \in A} \|b^{(e)} - a\|_2$ (or $\|b^{(e)}\|_2$ if $A = \emptyset$). Insert the top $q = \max(1, \lfloor E/2 \rfloor)$ behaviors by n_e into A . If $|A| > M$, remove the oldest entries.

7.7 Optional deterministic spawn/merge rules

These are optional (useful if you allow more than E internal eyes and then select the best E at output):

- **Spawn:** if $\text{Disagree}(x) > \delta_{\text{spawn}}$ for T_{spawn} consecutive refinement steps, spawn a new eye with predetermined parameter perturbation and temperature τ_{new} .
- **Merge/retire:** if

$$\frac{1}{N} \sum_{k=1}^N \text{JS}(p_k^{(e)}, p_k^{(e')}) < \delta_{\text{merge}},$$

retire the eye with lower proxy task score.

8 Deterministic Reconstruction of 3D Coordinates

Each eye yields expected distances for each pair (a_k, b_k) , then solves a weighted distance-geometry problem deterministically.

8.1 Expected distance per die

For eye e and die k :

$$\hat{d}_k^{(e)}(x) = \sum_{f=1}^F p_{k,f}^{(e)}(x) \bar{d}_f.$$

This choice avoids randomness and ensures determinism.

8.2 Weighted distance-geometry objective

Let $X^{(e)} = [x_1^{(e)}, \dots, x_L^{(e)}]^T \in \mathbb{R}^{L \times 3}$ with $x_r^{(e)} \in \mathbb{R}^3$. Define:

$$\begin{aligned} J^{(e)}(X) = & \sum_{k=1}^N w_k \left(\|x_{a_k} - x_{b_k}\|_2 - \hat{d}_k^{(e)} \right)^2 \\ & + \lambda_{\text{chain}} \sum_{r=1}^{L-1} (\|x_{r+1} - x_r\|_2 - d_{\text{chain}})^2 + \lambda_{\text{smooth}} \sum_{r=2}^{L-1} \|x_{r-1} - 2x_r + x_{r+1}\|_2^2, \end{aligned}$$

where $w_k \geq 0$ are fixed weights, d_{chain} is a fixed target step length, and $\lambda_{\text{chain}}, \lambda_{\text{smooth}}$ are fixed constants.

8.3 Deterministic optimization

Initialize each eye with a fixed helical perturbation:

$$x_r^{(e)}(0) = [r d_{\text{chain}}, \varepsilon \cos(\omega r + \phi_e), \varepsilon \sin(\omega r + \phi_e)],$$

where ε, ω are constants and $\phi_e = 2\pi(e-1)/E$.

Perform exactly T steps of fixed-step gradient descent with step size η :

$$X(t+1) = X(t) - \eta \nabla_X J^{(e)}(X(t)), \quad t = 0, \dots, T-1.$$

No early stopping; output $X^{(e)} := X(T)$.

9 How Spider-Eyes Supports RNA 3D Folding

RNA folding is multimodal: multiple conformations may satisfy local constraints. A single predictor often collapses to one mode.

Spider-Eyes supports folding by:

1. **Exploration via multiple projections.** Different eyes (different gates/temperatures) induce different expected-distance sets $\{\hat{d}_k^{(e)}\}$, leading to different reconstructed minima.
2. **Coherence via cross-eye agreement.** MoM disagreement controls pathological divergence among eyes.
3. **Diversity via novelty pressure.** Archive-based novelty discourages collapse to identical solutions.
4. **Direct alignment to five-candidate scoring.** Outputting five structured alternatives fits evaluation that rewards the best-of-five.

10 Determinism Checklist

To guarantee deterministic behavior (up to floating-point roundoff), fix:

- All scalar constants: E, F, D, G, C, h, H , bin edges $\{b_f\}$, $d_{\max}$, d_{chain} , λ_{chain} , λ_{smooth} , T , η , ε , ω , α, β, γ , archive capacity M , and any ϵ used for stability.
- All parameter tensors for every eye: $A^{(e)}$, $W_{c,j}^{(e)}$, $b_{c,j}^{(e)}$, $w_{k,f}^{(e)}$, $b_{k,f}^{(e)}$, and τ_e .
- All tie-breakers (e.g. if equal logits, choose smallest face index).
- All numeric conventions (log base for KL/JS, softmax definition).

11 End-to-End Deterministic Algorithm (Summary)

Given sequence s :

1. Build u_r and compute global context x .
2. Construct pair set P and distance bins.
3. For each eye $e = 1, \dots, 5$:
 - (a) Compute mosaic $m^{(e)}(x)$.
 - (b) Compute dice distributions $p_k^{(e)}(x)$ for all k .
 - (c) Compute expected distances $\hat{d}_k^{(e)}(x)$.
 - (d) Reconstruct $X^{(e)} \in \mathbb{R}^{L \times 3}$ by minimizing $J^{(e)}$ via fixed-step gradient descent.
4. Compute MoM metrics $\text{Disagree}(x)$, $\text{Novel}(x)$, and optional $\text{Task}(x)$ (diagnostic or optional control).
5. Output all five coordinate sets $\{X^{(e)}\}_{e=1}^5$ in the required submission format.